GEOG 4/5/7 9073: Environmental Analysis in R

Week 02.01: Data structures and programmatic thinking

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Today's schedule

- Open discussion
- Data structures
- Exercises

Anything to discuss? Questions?

Your homework for today

(setup your computing environment, review chapters 1-4 in *R for Data Science*, come to class with 2 questions about R, geospatial programming in general, or this course)

Find a *different* buddy

- Share:
  - How you found the setup process to go (e.g., did you run into any issues?)
  - What you thought of Chapters 1-4
  - Your questions

Report out (to the whiteboard!!!)

Bringing everyone along together

- One of our challenges in this kind of course
- Spatial analysis is not programming…
- …and programming is not spatial analysis

Some handy dandy operations on vectors

First, create a vector

```
x <- seq(1:20) # what does this do? How would you know?

# alterative method... seq is an "overloaded" function <- what does this mean?
x <- seq(1, 20, 1)
```

Operations

```
sum(x)
mean(x)
median(x)
sd(x)
length(x)
```

Data types

Table 2.1 Data type, tests and conversion functions

Type	Test	Conversion
character	`is.character`	`as.character`
complex	`is.complex`	`as.complex`
double	`is.double`	`as.double`
expression	`is.expression`	`as.expression`
integer	`is.integer`	`as.integer`
list	`is.list`	`as.list`
logical	`is.logical`	`as.logical`
numeric	`is.numeric`	`as.numeric`
single	`is.single`	`as.single`
raw	`is.raw`	`as.raw`

Factors can be a pain

- What's a "factor" according to your book?
- What are some key properties of factors?
    - Ordering
    - Levels

Interrogating types

the typeof() function

```
typeof(8675309)
typeof(integer(8675309))

typeof(TRUE)

typeof("banana")

typeof(rep(1, 10))

typeof(list(1, 3, 4, "orange"))
```

Let's look more closely at data frames and tibbles

From the course GitHub page, get "oh_counties_DP2020.csv" (it's in the data folder)

(https://github.com/pjbitterman/KSU_spatial_data_sci_R)

```
library(tidyverse) #get the helper functions

# read the data
mydf <- read_csv("./data/ohio/oh_counties_DP2020.csv)

# look at it
mydf
```

What do you see?

The value of exploratory data analysis (EDA)

- When you first get new data, it's a good idea to look at it before starting work
- Many ways of doing so... like what?

```
summary(mydf) # what do you get?
# How many observations does your data have?
nrow(mydf)
# and attibutes?
ncol(mydf)

# an easier way to look at attributes
glimpse(mydf)

# access a single attribute
mydf$poptotal # Total population
summary(mydf$poptotal)
hist(mydf$poptotal)
```

Subsetting your data

- Often you need to filter your data such that only those observations meeting certain criteria are retained (or removed)

```
# requires dplyr/tidyverse
dplyr::filter(mydf, poptotal > 50000 & medianage < 40)
```

Another way to write that function

```
mydf %>% dplyr::filter(., poptotal > 50000 & medianage < 40)
```

What's the `%>%` and how does it work?

The pipe `(%>%)`

- from magrittr package
- essentially says "take what's on the left and pass it to the right"
- R assumes you want to pipe to the first argument of the right-hand function, but...
- you can explicitly place the output of the pipe using a `.` on the right-hand side

```
mydf %>% dplyr::filter(., poptotal > 50000 & medianage < 40)
```

But what's the point? (to the whiteboard!!!)

PLOTTING

Why do we plot our data?

(I know it's a bit of a silly question, just go with it)

Many methods, we'll focus on two (but really just one)

1. "base R"
2. ggplot

Setup

Required packages

```
library(tidyverse)
library(ggplot2)
```

remind yourself what the data look like

```
mydf
glimpse(mydf) # a useful function
```

Some simple base R plots

the "plot" function can be used in multiple ways

```
# scatter plot
# should be highly correlated
plot(mydf$poptotal, mydf$Hhtotal)
```

What do you see?

```
# scatter plot #2
plot(mydf$medianage, mydf$Hhtotal)
```

Let's try a histogram

```
# histogram
hist(mydf$Hhtotal)
```

and exert a bit more control

```
# change the number of breaks
hist(mydf$Hhtotal, breaks = 20)
```

ggplot : "the grammar of graphics"

Let's build a plot step-by-step

the setup

```
# the initial call
ggplot(mydf, aes(x = poptotal, y = medianage))
```

What happened?

Let's add a geom

Wait, what's a "geom"???

```
ggplot(mydf, aes(x = poptotal, y = medianage)) +
  geom_point()
```

Note the "+" ...it's a very different notation

and we can modify the points:

```
ggplot(mydf, aes(x = poptotal, y = medianage)) +
  geom_point(colour = "blue")
```

And alter the theme

```
ggplot(mydf, aes(x = poptotal, y = medianage)) +
  geom_point(colour = "blue") +
  theme_minimal()
```

Try a different one! How would you know what options there are for themes?

Add some labels

```
ggplot(mydf, aes(x = poptotal, y = medianage)) +
  geom_point(colour = "blue") +
  theme_minimal() +
  labs(x = "Total Population", y = "Median Age")
```

...and add a title

```
# and give it a title
ggplot(mydf, aes(x = poptotal, y = medianage)) +
  geom_point(colour = "blue") +
  theme_minimal() +
  labs(x = "Total Population", y = "Median Age",
  title = "My First ggplot")
```

fit a line

don't just throw arbitrary models/fits on your data

```
ggplot(mydf, aes(x = poptotal, y = medianage)) +
  geom_point(colour = "blue") +
  geom_smooth(method = "glm", colour = "red") +
  theme_minimal() +
  labs(x = "Total Population", y = "Median Age",
  title = "My First ggplot")
```

Let's try something different

Using categorical data

(first, we have to make some categories) - let's walk through this code

```
mydf2 <- mydf %>% mutate(sizeCategory = ifelse(poptotal > 100000, "big", "small"))
```

check your work:

```
summary(mydf2$sizeCategory) ### What happened?

# turn them into factors to count them
summary(as.factor(mydf2$sizeCategory))
```

A first example

Let's break it down:

```
ggplot(mydf2, aes(x = poptotal, y = medianage)) +
  geom_point(aes(shape = sizeCategory, colour = sizeCategory), size = 3) +
  theme_minimal() +
  labs(x = "Total Population", y = "Median Age",
       title = "My formatted ggplot")
```

what happened?

One more example:

a pipe with a boxplot

```
mydf2 %>% ggplot(., aes(x = sizeCategory, y = medianage)) +
  geom_boxplot(aes(fill = sizeCategory)) +
  theme_minimal() +
  labs(x = "Categorical size",
       y = "Median Age",
       title = "I made a boxplot",
       subtitle = "...it's handy for comparing groups")
```

If there's time, try it yourself:

using ggplot:

- Make a histogram
- Try a barplot...
  - subset of the counties (however you want)
  - counties on the x-axis, number of vacant units on the y-axis

Review and next class

- Any questions?
- This week’s readings/tasks:
  - Chapter 2 in textbook
  - Continue to review Hadley's book/site
  - Practice on your own